

Linear Regression using OLS

(Ordinary Least Square) — Full Derivation, Detailed Notes (Hinglish)

1. Basic Idea: OLS kya hai?

Linear Regression me hum ek best-fit line dhoondte hain jo data points ke saath sabse achhi tarah match kare. Us line ko dhoondne ka process 'Optimization' kehlata hai, aur optimization ke andar hum ek 'Loss Function' (ya Cost Function) ko minimize karte hain.

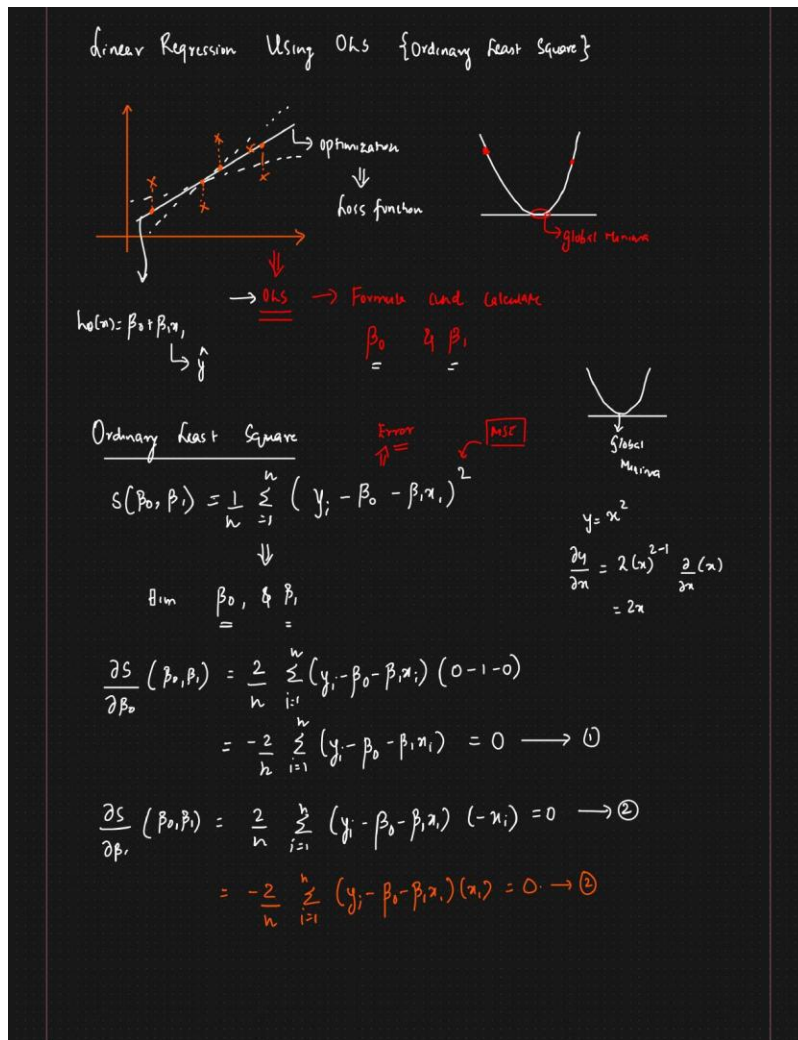
Jaise humne pehle dekha tha, cost function $J(\theta)$ ka graph ek 'U' shape (bowl) me hota hai, jiska sabse neeche wala point Global Minima kehlata hai — yehi wo point hai jaha error sabse kam hota hai, aur yehi hume best line ke parameters deta hai.

OLS (Ordinary Least Square) ek method hai jisse hum is best-fit line ke exact parameters — Intercept (β_0) aur Coefficient/Slope (β_1) — ko Formula banake seedha Calculate kar sakte hain (bina baar-baar iterate kiye, jaisa Gradient Descent me hota hai).

Hypothesis Function

$$h\theta(x) = \beta_0 + \beta_1 x \rightarrow \hat{y} \text{ (predicted value)}$$

- $h\theta(x)$ = model ka hypothesis/prediction function — ye line ka equation hai.
- β_0 = Intercept — line y-axis ko kaha cut karti hai (jab $x = 0$ ho).
- β_1 = Coefficient / Slope — line kitni tez badhti/ghatti hai jab x badhta hai.
- $\hat{y}$ (**y-hat**) = model ki predicted value — hypothesis function ka output.



Original Notes — Page 1: OLS Intro, Hypothesis Function, Cost Function $S(\beta_0, \beta_1)$

2. Ordinary Least Square — Cost Function

OLS ka cost function bilkul MSE (Mean Squared Error) jaisa hi hota hai — bas isme y -hat ($\hat{y}$) ki jagah seedha $(\beta_0 + \beta_1 x_i)$ likh dete hain, kyunki yehi to model ka prediction hai.

$$S(\beta_0, \beta_1) = (1/n) \times \sum (y_i - \beta_0 - \beta_1 x_i)^2 \quad [i = 1 \text{ to } n]$$

- $S(\beta_0, \beta_1)$ = cost function — ye batata hai ki diye gaye β_0 aur β_1 ke sath model ka total error kitna hai.
- n = total data points
- y_i = i -th point ki actual value
- $\beta_0 + \beta_1 x_i$ = i -th point ke liye model ki predicted value ($\hat{y}_i$)
- $(y_i - \beta_0 - \beta_1 x_i)^2$ = us point ka squared error

Hume aisa β_0 aur β_1 dhoondna hai jisse $S(\beta_0, \beta_1)$ sabse minimum ho — yani jaha ye function apne Global Minima par pahunchne.

Minima kaise dhoondte hain? — Derivative = 0

Kisi bhi function ka minima (ya maxima) dhoondne ka standard tarika hai: uska derivative (slope) nikalo aur use 0 ke barabar rakh do — kyunki bowl ke sabse neeche wale point par curve ka slope hamesha 0 hota hai (flat point).

Ek simple example se samjho — agar $y = x^2$, to power rule se derivative hota hai:

$$dy/dx = 2(x)^{(2-1)} = 2x$$

Bilkul isi tarah, hume apne cost function $S(\beta_0, \beta_1)$ ka Partial Derivative — pehle β_0 ke respect me, phir β_1 ke respect me — nikalna hoga, aur dono ko 0 ke barabar rakhna hoga. (Partial derivative isliye kyunki S do variables β_0 aur β_1 par depend karta hai — hum ek waqt me sirf ek variable ke respect me differentiate karte hain, doosre ko constant maan ke).

3. Equation 1: $\partial S/\partial \beta_0 = 0$ (Intercept ke liye)

Sabse pehle $S(\beta_0, \beta_1)$ ko β_0 ke respect me partial differentiate karte hain. Chain rule use hoti hai: andar wale bracket $(y_i - \beta_0 - \beta_1 x_i)$ ka β_0 ke respect me derivative sirf '-1' hoga (kyunki y_i aur $\beta_1 x_i$, β_0 ke respect me constant hain).

$$\partial S/\partial \beta_0 = (2/n) \sum (y_i - \beta_0 - \beta_1 x_i)(0 - 1 - 0) = -(2/n) \sum (y_i - \beta_0 - \beta_1 x_i) = 0$$

Chunki $(2/n)$ sirf ek constant hai aur poore equation ko 0 karne ke liye zaroori nahi ki ye bhi 0 ho, hum ise dono taraf se cancel/hata sakte hain:

$$\sum (y_i - \beta_0 - \beta_1 x_i) = 0 \rightarrow \text{Equation (1)}$$

Equation (1) ko solve karna:

Summation ko individually expand karte hain — har term par alag se $\sum$ lagate hain:

$$-\sum y_i + n \cdot \beta_0 + \beta_1 \cdot \sum x_i = 0$$

Yaha ' $\sum \beta_0$ ' ban gaya ' $n \times \beta_0$ ' — kyunki β_0 ek constant hai aur use n baar add kiya ja raha hai ($i=1$ se n tak).

Ab dono taraf rearrange karte hain, β_0 ko akela chhodte hain:

$$n \cdot \beta_0 + \beta_1 \cdot \sum x_i = \sum y_i \rightarrow \beta_0 = (\sum y_i)/n - \beta_1 \cdot (\sum x_i)/n$$

Yaha $(\sum y_i)/n$ asal me y ka Mean (average) hai — jise $\bar{y}$ (y-bar) likhte hain. Isi tarah $(\sum x_i)/n$, x ka Mean hai — $\bar{x}$ (x-bar). Isse formula bahut simple ban jata hai:

$$\text{Intercept: } \beta_0 = \bar{y} - \beta_1 \bar{x}$$

Ye formula bahut important hai — isse pata chalta hai ki Intercept (β_0) seedha data ke means ($\bar{y}$, $\bar{x}$) aur Slope (β_1) se calculate ho sakta hai.

4. Equation 2: $\partial S/\partial \beta_1 = 0$ (Coefficient/Slope ke liye)

Ab $S(\beta_0, \beta_1)$ ko β_1 ke respect me partial differentiate karte hain. Is baar andar wale bracket ka β_1 ke respect me derivative ' $-x_i$ ' hoga (kyunki β_1 ke sath x_i multiply hai).

$$\begin{aligned} \partial S/\partial \beta_1 &= (2/n) \sum (y_i - \beta_0 - \beta_1 x_i)(-x_i) = 0 \\ &-(2/n) \sum (y_i - \beta_0 - \beta_1 x_i)(x_i) = 0 \rightarrow \text{Equation (2)} \end{aligned}$$

$(2/n)$ ko dono taraf se cancel karke aur andar ka bracket x_i se multiply karke expand karte hain:

$$\sum x_i y_i - \beta_0 \cdot \sum x_i - \beta_1 \cdot \sum x_i^2 = 0$$

$$\begin{aligned}
 \varepsilon_y &\rightarrow 0 \\
 -\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) &= 0 \\
 \Downarrow \\
 -\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) &= 0 \\
 -\sum_{i=1}^n y_i + n\beta_0 + \beta_1 \sum_{i=1}^n x_i &= 0 \\
 n\beta_0 + \beta_1 \sum_{i=1}^n x_i &= \sum_{i=1}^n y_i \\
 \beta_0 &= \frac{\sum_{i=1}^n y_i}{n} - \beta_1 \frac{\sum_{i=1}^n x_i}{n} \\
 \text{Intercept} \rightarrow \boxed{\beta_0 = \bar{y} - \beta_1 \bar{x}} \\
 \boxed{\varepsilon_y 2^{\circ}} \\
 -\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)(x_i) &= 0 \\
 \Downarrow \\
 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)(x_i) &= 0 \\
 \Downarrow \\
 \sum_{i=1}^n x_i y_i - \beta_0 \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 &= 0
 \end{aligned}$$

Original Notes — Page 2: Equation (1) solve karke β_0 (Intercept) nikalna, Equation (2) setup

5. β_0 ko Equation (2) me Substitute karna

Ab jo Equation (1) se $\beta_0 = \bar{y} - \beta_1 \bar{x}$ mila tha, use Equation (2) me daal dete hain, taaki sirf β_1 hi unknown reh jaye:

$$\sum x_i y_i - (\bar{y} - \beta_1 \bar{x}) \cdot \sum x_i - \beta_1 \cdot \sum x_i^2 = 0$$

Bracket ko expand karte hain:

$$\sum x_i y_i - \bar{y} \cdot \sum x_i + \beta_1 \cdot \bar{x} \cdot \sum x_i - \beta_1 \cdot \sum x_i^2 = 0$$

Ab β_1 ko common factor ki tarah bahar nikaal ke, terms ko is tarah rearrange karte hain ki ek side sirf β_1 wale terms ho aur doosri side baaki sab:

$$\sum (x_i y_i - \bar{x} y_i) = \beta_1 \cdot \sum (x_i^2 - \bar{x} x_i)$$

Ye expression algebra ki madad se ek bahut hi clean, well-known form me convert ho jaati hai — jaha $\sum x_i y_i - n \bar{x} \bar{y}$ wagera terms ko $(x_i - \bar{x})$ aur $(y_i - \bar{y})$ ke roop me dobara likha ja sakta hai (ye ek standard algebraic identity hai, jise 'deviation form' kehte hain):

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = \beta_1 \cdot \sum (x_i - \bar{x})^2$$

Isse aakhir me β_1 (Slope/Coefficient) ka final formula milta hai:

$$\text{Coefficient: } \beta_1 = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \Sigma(x_i - \bar{x})^2$$

Note: Original hand-written notes me is aakhri step par derivation thoda shortcut le raha hai aur numerator/denominator me $(x_i - \bar{x})$ wala factor likhna reh gaya hai (sirf $\Sigma(y_i - \bar{y})$ aur $\Sigma(x_i - \bar{x})$ dikh raha hai). Lekin mathematically sahi aur standard OLS formula wahi hai jo upar diya gaya hai — Numerator me $(x_i - \bar{x})(y_i - \bar{y})$ ka product, aur Denominator me $(x_i - \bar{x})^2$ ka sum hota hai. Ye formula 'covariance of x,y divided by variance of x' jaisi bhi likha ja sakta hai.

$$\begin{aligned} \sum_{i=1}^n (x_i y_i - \beta_0 x_i - \beta_1 x_i^2) &= 0 \\ \downarrow \\ \text{Replace } \beta_0 &= \bar{y} - \beta_1 \bar{x} \\ \sum_{i=1}^n (x_i y_i - (\bar{y} - \beta_1 \bar{x}) x_i - \beta_1 x_i^2) &= 0 \\ \sum_{i=1}^n (x_i y_i - x_i \bar{y} + \beta_1 \bar{x} x_i - \beta_1 x_i^2) &= 0 \\ \sum_{i=1}^n (y_i - \bar{y} + \beta_1 \bar{x} - \beta_1 x_i) x_i &= 0 \\ \sum_{i=1}^n [(y_i - \bar{y}) + \beta_1 (\bar{x} - x_i)] x_i &= 0 \\ \sum_{i=1}^n (y_i - \bar{y}) x_i + \sum_{i=1}^n \beta_1 (\bar{x} - x_i) x_i &= 0 \\ \sum_{i=1}^n \beta_1 (\bar{x} - x_i) x_i &= - \sum_{i=1}^n (y_i - \bar{y}) x_i \\ \beta_1 &= \frac{- \sum_{i=1}^n (y_i - \bar{y}) x_i}{\sum_{i=1}^n (\bar{x} - x_i) x_i} \\ \beta_1 &= \frac{- \sum_{i=1}^n (y_i - \bar{y})}{\sum_{i=1}^n (\bar{x} - x_i)} \\ \text{Coefficient } \Rightarrow \beta_1 &= \frac{\sum_{i=1}^n (y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})} \end{aligned}$$

Original Notes — Page 3: β_0 ko Equation (2) me substitute karke β_1 (Coefficient) ka formula derive karna

6. Final OLS Formulas (Summary)

$$\beta_1 \text{ (Coefficient)} = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \Sigma(x_i - \bar{x})^2$$

$$\beta_0 \text{ (Intercept)} = \bar{y} - \beta_1 \cdot \bar{x}$$

In dono formulas se hum bina kisi iteration/loop ke, seedha ek formula lagakar best-fit line ke Intercept aur Coefficient nikaal sakte hain — isi wajah se OLS ko 'Closed-Form Solution' bhi kaha jata hai.

Practical tarika (Table Method)

Real data par calculate karne ke liye, aksar ek table banayi jaati hai jisme har row me x , y , $(y_i - \bar{y})$, aur $(x_i - \bar{x})$ columns hote hain. In columns ko multiply/square karke unke sums nikaal lete hain, aur neeche means $(\bar{x}, \bar{y})$ bhi calculate kar lete hain — phir seedha formula me values daal dete hain.

- **Column X, Y** = raw data points
- $(y_i - \bar{y})$ = har y value ka apne mean se difference
- $(x_i - \bar{x})$ = har x value ka apne mean se difference
- β_1 = in columns ko use karke final coefficient calculate karna
- $\beta_0 = (\bar{y} - \beta_1 \bar{x})$ = phir coefficient use karke intercept nikalna

Intercept
↓
 $\beta_0 = \bar{y} - \beta_1 \bar{x}$

OLS ↑
coefficient

coefficient Intercept
↓ ↓
 β_1 $\beta_0 = \bar{y} - \beta_1 \bar{x}$

x	y	$(y_i - \bar{y})$	$(x_i - \bar{x})$
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
$\bar{x}$	$\bar{y}$		

OLS $\approx$ Linear Regression (sklearn)

Original Notes — Page 4: Final Intercept/Coefficient formulas aur Table Method

Notes ke last line me likha hai: 'OLS $\approx$ Linear Regression (sklearn)' — matlab jab hum Python ki sklearn library me `LinearRegression().fit()` call karte hain, to background me wahi ye OLS wale formulas (ya inke equivalent matrix-based version) use hoke β_0 aur β_1 calculate karte hain.